R Programming Tutorial (Part 3)

Prepared by: Chan Weng Howe

UTM UNIVERSITI TEKNOLOGI MALAYSIA

Outline

- Hypothesis Testing in R (2 samples)
  - Two proportions
  - Two means: independent samples
    - Variance / Standard deviations Known
    - Variance / Standard deviations Unknown
      - Equal Variance
      - Unequal Variance

Two Proportions

Example

- Below are the sample data and summary statistics:

	Group 1	Group 2
	Subjects Given $1 Bill	Subjects Given 4 Quarters
Spent the money	$x_1=12$	$x_2=27$
Subjects in group	$n_1=46$	$n_2=43$

$$\hat{p}_1 = \frac{12}{46} \qquad \hat{p}_2 = \frac{27}{43}$$

$$\bar{p} = \frac{12+27}{46+43} = 0.438202$$

+n2)

/n1)+(pbar*qbar

pha)

.64485362695147"
8739527429506"

Two Proportions

WHAT WE KNOW?

- Two samples:
  - $x_1=12$, $x_2=27$
  - $n_1=46$, $n_2=43$
- Significance level: 0.05

WHAT WE WANT TO FIND?

$$H_0 : p_1 = p_2$$

$$H_1 : p_1 < p_2$$

$$\bar{p} = \frac{x_1 + x_2}{n_1 + n_2} \qquad \bar{q} = 1 - \bar{p}$$

$$z = \frac{(\hat{p}_1 - \hat{p}_2) - (p_1 - p_2)}{\sqrt{\frac{\bar{p}\bar{q}}{n_1} + \frac{\bar{p}\bar{q}}{n_2}}}$$

```
x1 = 12
x2 = 27
n1 = 46
n2 = 43

phat1 <- x1/n1
phat2 <- x2/n2

pbar = (x1+x2)/(n1+n2)
qbar = 1-pbar

z = ((phat1-phat2)-
0)/sqrt((pbar*qbar/n1)+(pbar*qbar
/n2))

alpha = 0.05
z.alpha = qnorm(alpha)
```

```
[1] "Critical Value: -1.64485362695147"
[1] "Z statistics: -3.48739527429506"
```

Two Proportions

Exercise

- Time magazine reported the result of a telephone poll of 800 adult Americans. The question posed of the Americans who were surveyed was: "Should the federal tax on cigarettes be raised to pay for health care reform?" The results of the survey were:

Non-smoker	Smoker
N_1=605	N_2=195
Y1=351 said yes	Y2=41 said yes

- Is there sufficient evidence at the α = 0.05 level, say, to conclude that the two populations — smokers and non-smokers — differ significantly with respect to their opinions?

+n2)

/n1)+(pbar*qbar

pha/2)

5996398454005"
22953986016"

Two Proportions

WHAT WE KNOW?

- Two samples:
  - x_1=351, x_2=41
  - n_1=605, n_2=195
- Significance level: 0.05

WHAT WE WANT TO FIND?

$$H_0 : p_1 = p_2$$

$$H_1 : p_1 \neq p_2$$

$$\bar{p} = \frac{x_1 + x_2}{n_1 + n_2} \qquad \bar{q} = 1 - \bar{p}$$

$$z = \frac{(\hat{p}_1 - \hat{p}_2) - (p_1 - p_2)}{\sqrt{\dfrac{\bar{p}\bar{q}}{n_1} + \dfrac{\bar{p}\bar{q}}{n_2}}}$$

```
x1 = 351
x2 = 195
n1 = 605
n2 = 195

phat1 <- x1/n1
phat2 <- x2/n2

pbar = (x1+x2)/(n1+n2)
qbar = 1-pbar

z = ((phat1-phat2)-
0)/sqrt((pbar*qbar/n1)+(pbar*qbar
/n2))

alpha = 0.05
z.alpha = qnorm(alpha/2)
```

```
[1] "Critical Value: -1.95996398454005"
[1] "Z statistics: -10.9522953986016"
```

Two Means, independent samples, known σ^2

WHAT WE KNOW?

Example

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A product developer is interested in reducing the drying time of a primer paint. Two formulations of the paint are tested; formulation 1 is the standard chemistry, and formulation 2 has a new drying ingredient that should reduce the drying time. From experience, it is known that the standard deviation of drying time is 8 minutes, and this inherent variability should be unaffected by the addition of the new ingredient. Ten specimens are painted with formulation 1, and another ten specimens are painted with formulation 2; the 20 specimens are painted in random order and normally distributed. The two sample average drying times are $\bar{x}_1$ = 121 minutes and $\bar{x}_2$ = 112 minutes, respectively. What conclusions can the product developer draw about the effectiveness of the new ingredient, using α = 0.05?

```
gma2^2
gma2^2

ar2-
ance1/n1)+(variance

m(alpha,
SE)
```

```
e: 1.64485362695147"
 2.51557647468726"
```

Two Means, independent samples, known σ^2

WHAT WE KNOW?

- Two samples:
  - $\sigma_1 = \sigma_2 = 8$
  - $n_1=10$, $n_2=10$
  - $\bar{x}_1 = 121$, $\bar{x}_2 = 112$

WHAT WE WANT TO FIND?

$$H_0 : \mu_1 = \mu_2$$

$$H_1 : \mu_1 > \mu_2$$

$$z_0 = \frac{\bar{x}_1 - \bar{x}_2 - 0}{\sqrt{\frac{\sigma_1^2}{n1} + \frac{\sigma_2^2}{n2}}}$$

```
xbar1 = 121
xbar2 = 112

n1 = 10
n2 = 10

sigma1 = 8
sigma2 = 8

variance1 = sigma2^2
variance2 = sigma2^2

z = ((xbar1-xbar2-
0)/(sqrt((variance1/n1)+(variance
2/n2))))

alpha = 0.05
z.alpha = qnorm(alpha,
lower.tail=FALSE)
```

```
[1] "Critical Value: 1.64485362695147"
[1] "Z statistics: 2.51557647468726"
```

Two Means, independent samples, unknown σ^2 (Case: equal σ^2)

WHAT WE KNOW?

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Case 1 : $\sigma_1^2 = \sigma_2^2 = \sigma^2$

- **Example**

Two catalysts are being analyzed to determine how they affect the mean yield of a chemical process. Specifically, catalyst 1 is currently in use, but catalyst 2 is acceptable. Since catalyst 2 is cheaper, it should be adopted, providing it does not change the process yield. A test is run in the pilot plant and results in the data shown in Table 5.1. Is there any difference between the mean yields? Use α = 0.05, and assume equal variances.


```
sigma2^2
sigma2^2

-xbar2-
ariance1/n1)+(variance

5
norm(alpha,
FALSE)
```

```
.1447866879178 2.1447866879178"
5359086434619"
```

Two Means, independent samples, unknown σ² (Case: equal σ²)

WHAT WE KNOW?

- Two samples:
  - Two sets of data
  - Significance: 0.05

WHAT WE WANT TO FIND?

$$H_0 : \mu_1 = \mu_2$$
$$H_1 : \mu_1 \neq \mu_2$$

$$S_p^2 = \frac{(n_1 - 1)S_1^2 + (n_2 - 1)S_2^2}{n_1 + n_2 - 2}$$

$$t_0 = \frac{\bar{x}_1 - \bar{x}_2 - 0}{S_p\sqrt{\frac{1}{n1} + \frac{1}{n2}}}$$

```
xbar1 = 121
xbar2 = 112

n1 = 10
n2 = 10

sigma1 = 8
sigma2 = 8

variance1 = sigma2^2
variance2 = sigma2^2

z = ((xbar1-xbar2-
0)/(sqrt((variance1/n1)+(variance
2/n2))))

alpha = 0.05
z.alpha = qnorm(alpha,
lower.tail=FALSE)
```

```
[1] "Critical Value: -2.1447866879178 2.1447866879178"
[1] "T statistics: -0.35359086434619"
```

Two Means, independent samples, unknown σ² (Case: unequal σ²)

Example

Arsenic concentration in public drinking water supplies is a potential health risk. An article in the *Arizona Republic* (Sunday, May 27, 2001) reported drinking water arsenic concentrations in parts per billion (ppb) for 10 metropolitan Phoenix communities and 10 communities in rural Arizona. The data is given as follows:


```
5
5
-xbar2-
1^2/n1)+(s2^2/n2)))
(s2^2/n2))^2/((((s1^2/
/n2)^2)/(n2-1)))
5
t(alpha/2, floor(v))
```

```
 -2.16036865646279"
[1] "T statistics: -2.7744169270369"
```

Two Means, independent samples, unknown σ^2 (Case: unequal σ^2)

WHAT WE KNOW?

- Two samples:
    - Two sets of data
    - $s_1 = 7.63$, $s_2 = 15.3$
    - $n_1=10$, $n_2=10$
    - $\bar{x}_1 = 12.5$, $\bar{x}_2 = 27.5$
    - Significance: 0.05

WHAT WE WANT TO FIND?

$$H_0 : \mu_1 = \mu_2$$
$$H_1 : \mu_1 \neq \mu_2$$

$$v = \frac{\left(\frac{S_1^2}{n1} + \frac{S_2^2}{n2}\right)^2}{\frac{\left(S_1^2/n1\right)^2}{n1-1} + \frac{\left(S_2^2/n2\right)^2}{n2-1}}.$$

$$T_0^* = \frac{\bar{X}_1 - \bar{X}_2 - \Delta_0}{\sqrt{\frac{S_1^2}{n1} + \frac{S_2^2}{n2}}}$$

```
xbar1 = 12.5
xbar2 = 27.5

s1 = 7.63
s2 = 15.3

n1 = 10
n2 = 10

t0 = (xbar1-xbar2-
0)/(sqrt((s1^2/n1)+(s2^2/n2)))

v =
((s1^2/n1)+(s2^2/n2))^2/((((s1^2/
n1)^2)/(n1-
1))+(((s2^2/n2)^2)/(n2-1)))

alpha = 0.05
t.alpha = qt(alpha/2, floor(v))

```

```
[1] "Critical Value: -2.16036865646279"
[1] "T statistics: -2.7744169270369"
```

THE END…